

(eliminating background noise, this is why it is recommended that you use a good array microphone for speech dictation, which does most of the noise cancellation work for the software). The system then needs to convert these “possible speech” elements into a series of numeric values which define the sounds in audio signal. Under Vista, these numeric values are then used to launch a search across three databases: an acoustic model, a lexicon, and a language model.

The acoustic model represents the sounds of a language (under Vista, you can train this module to better recognise your speech patterns). The lexicon is, of course, a comprehensive list of the words that form the language; it also offers pronunciation guides on each word. Now that the system knows what you are saying, it uses the language model to determine the best manner in which the different words you spoke need to be combined.

Human speech, of course, does not possess a precisely definable property—the manner in which we speak can vary through a day, depending upon mood, or health, or any number of factors. It is therefore important to collect as much data on the user’s voice as possible. To this end, Windows Vista offers to spider through the user’s documents and emails to better understand the writing style of the user and to create a better user model. The quality of speech recognition also depends upon how well the backend of the system refines the database; and eliminates poor matches.

4.6.1 Accessing Speech Recognition under Windows Vista

Before you can use this feature under Vista, you need to quickly take the program through your voice-train it to better suit your pronunciation and intonations. The training can be accessed from the Start menu—**All Programs > Accessories > Accessibility > Speech Recognition**. Or you can do it the modern way by just typing in “speech” in the Search field of the Start Menu and selecting the program. The tutorial will first take you through setting up your microphone and then takes you through an interactive tutorial that explains how to use speech recognition technology and helps you train the system to understand the your speech.